

Load Dataset

```
import pandas as pd

df = pd.read_csv("/content/marketing_and_sales_data_evaluate_lr.csv")
df.head()
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181



Next steps:

[Generate code with df](#)[New interactive sheet](#)

Explore dataset

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   TV               4562 non-null   float64
1   Radio            4568 non-null   float64
2   Social_Media     4566 non-null   float64
3   Sales            4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

Checking for missing values

```
df.isnull().sum()
```

	0
TV	10
Radio	4
Social_Media	6
Sales	6

```
dtype: int64
```

handle missing values by filling with mean and clean dataset properly

```
df_clean = df.copy()
```

```
for col in df_clean.columns:  
    df_clean[col] = df_clean[col].fillna(df_clean[col].mean())
```

verify cleaning worked

```
df_clean.isnull().sum()
```

```

      0
TV      0
Radio    0
Social_Media  0
Sales    0

```

dtype: int64

Data cleaning explanation The dataset contained missing values across all variables, with TV having the highest number of missing entries. Since the dataset is numerical and suitable for regression analysis, missing values were handled using mean imputation to preserve data integrity without losing observations.

```
df_clean.corr()
```

	TV	Radio	Social_Media	Sales
TV	1.000000	0.866885	0.527010	0.996652
Radio	0.866885	1.000000	0.606793	0.867369
Social_Media	0.527010	0.606793	1.000000	0.528121
Sales	0.996652	0.867369	0.528121	1.000000



Understanding the Data

```
df_clean.info()
df_clean.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   TV               4572 non-null   float64
1   Radio            4572 non-null   float64
2   Social_Media     4572 non-null   float64
3   Sales            4572 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

	TV	Radio	Social_Media	Sales
count	4572.000000	4572.000000	4572.000000	4572.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.096461	9.672723	2.211218	93.071947
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.546519	1.529879	112.397660
50%	53.000000	17.875181	3.057150	189.632530
75%	77.000000	25.647476	4.805832	272.414914
max	100.000000	48.871161	13.981662	364.079751

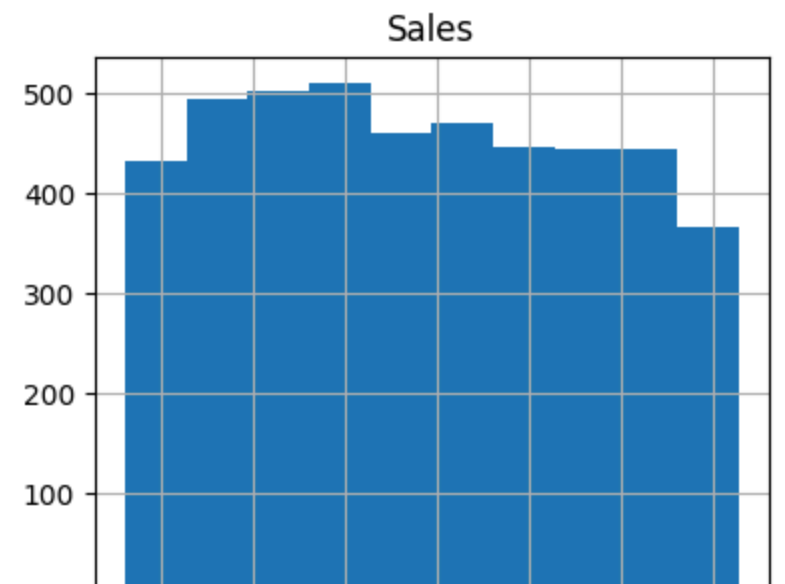
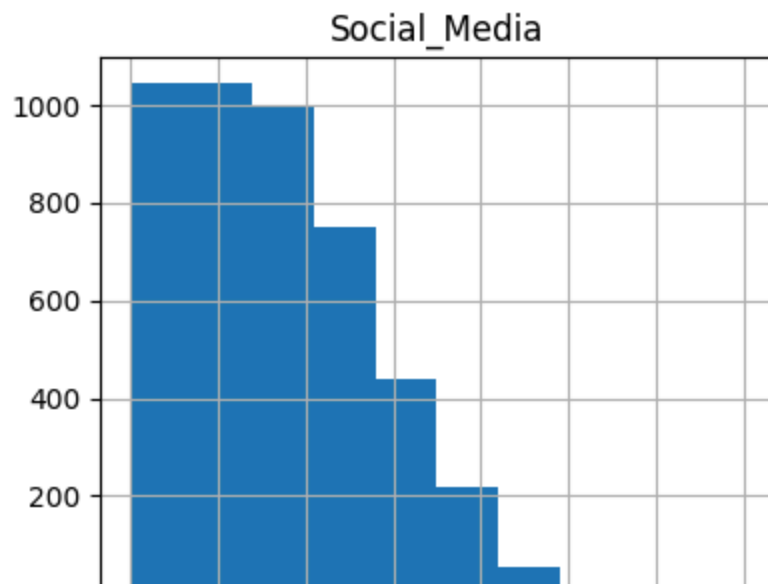
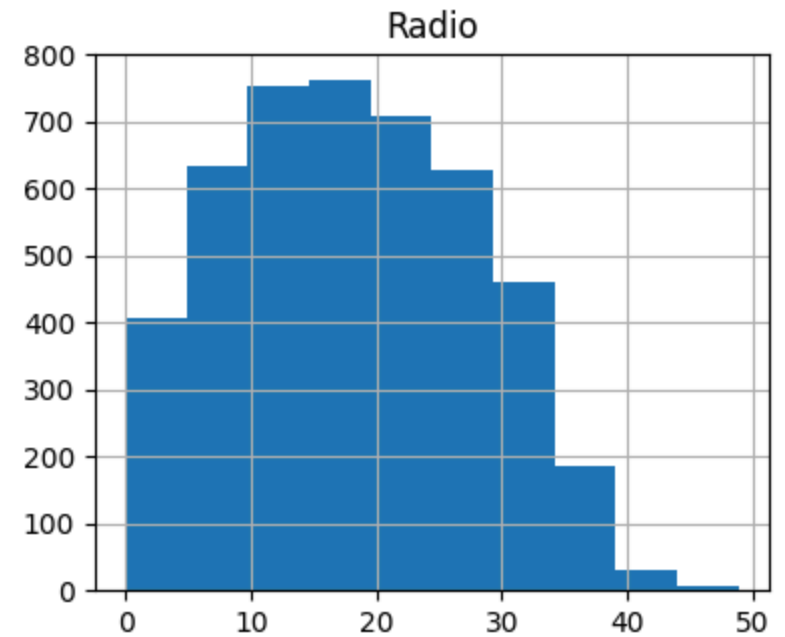
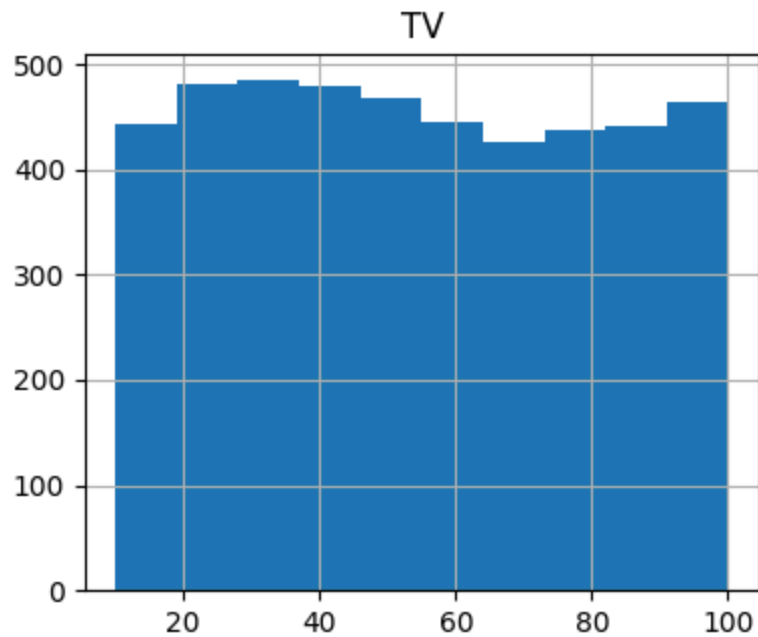
Performing exploratory data analysis (EDA) with visualizations

- Univariate Analysis
- Bivariate Analysis

Univariate Analysis

Histogram (distribution of each variable)

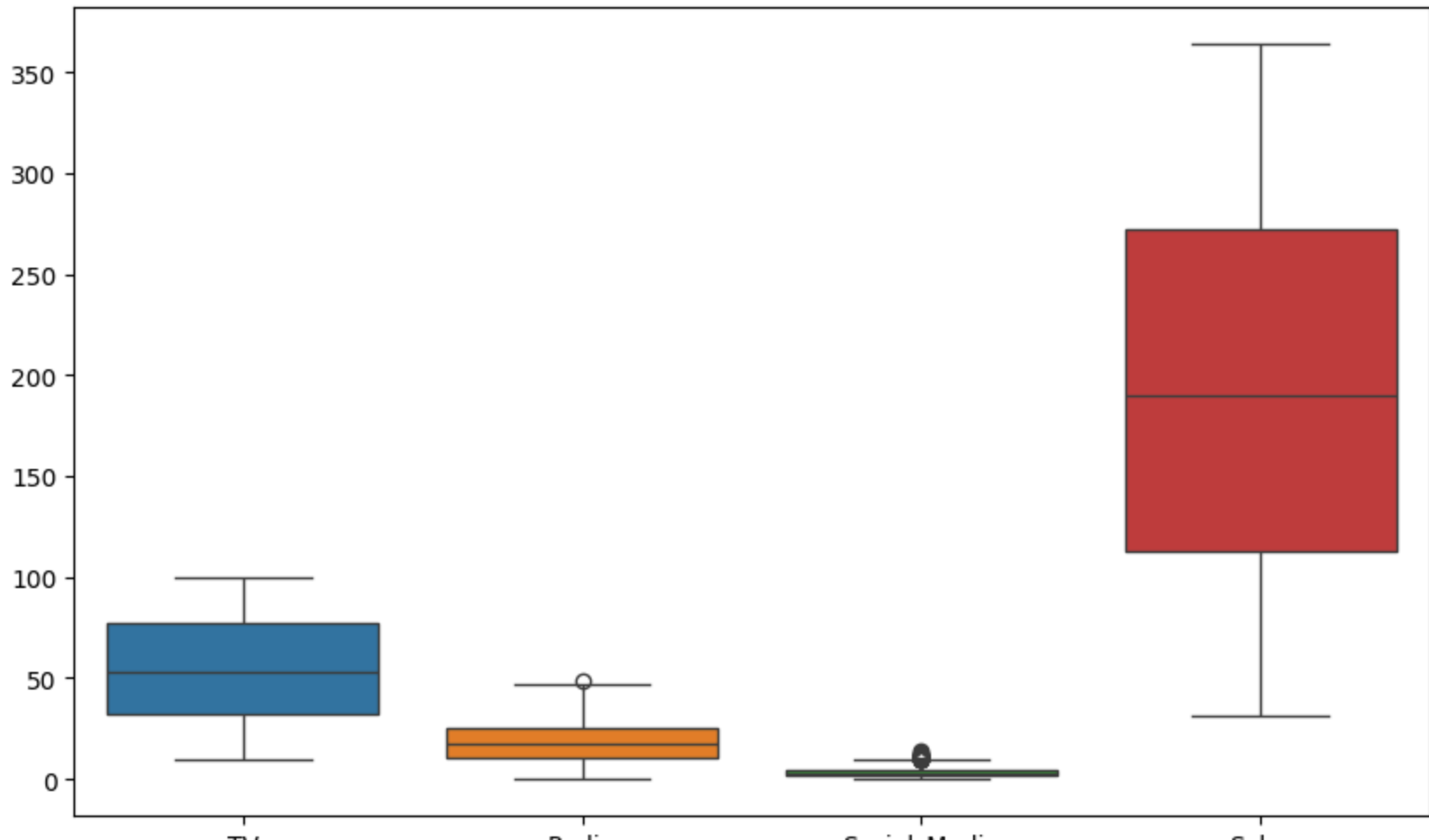
```
import matplotlib.pyplot as plt  
df_clean.hist(figsize=(10,8))  
plt.show()
```



Box Plot

```
import seaborn as sns

plt.figure(figsize=(10,6))
sns.boxplot(data=df_clean)
plt.show()
```



Interpretation of Boxplots What This Boxplot Shows for the following variables

- TV

- Radio
- Social media
- sales

Each box shows:

- Median (middle line)
- Spread (IQR) (box height)
- Outliers (dots)
- Range (whiskers)

Interpretation (Variable by Variable)

1. TV Median ≈ 50 Wide spread (values roughly $10 \rightarrow 100$)

Meaning:

TV Advertising spend varies a lot across campaigns Good variability $\rightarrow$ useful for regression No extreme outliers $\rightarrow$ clean variable

2. Radio Median ≈ 20 Smaller spread than TV Has a few outliers (dots near $\sim 45-50$)

Meaning:

Radio budget is more consistent A few campaigns spent unusually high $\rightarrow$ potential influence points

3. Social Media Median $\approx 3-4$ Very tight spread A couple of small outliers ($\sim 10-12$) Meaning:

Very low spend overall Little variation $\rightarrow$ might be a weak predictor in regression

4. Sales (VERY IMPORTANT) Median ≈ 190 Very wide spread ($\sim 30 \rightarrow 300$)

Meaning:

Sales vary a lot $\rightarrow$ good No obvious extreme outliers $\rightarrow$ data looks usable

Key Insights

TV advertising spend shows high variability, making it a strong candidate for predicting sales.

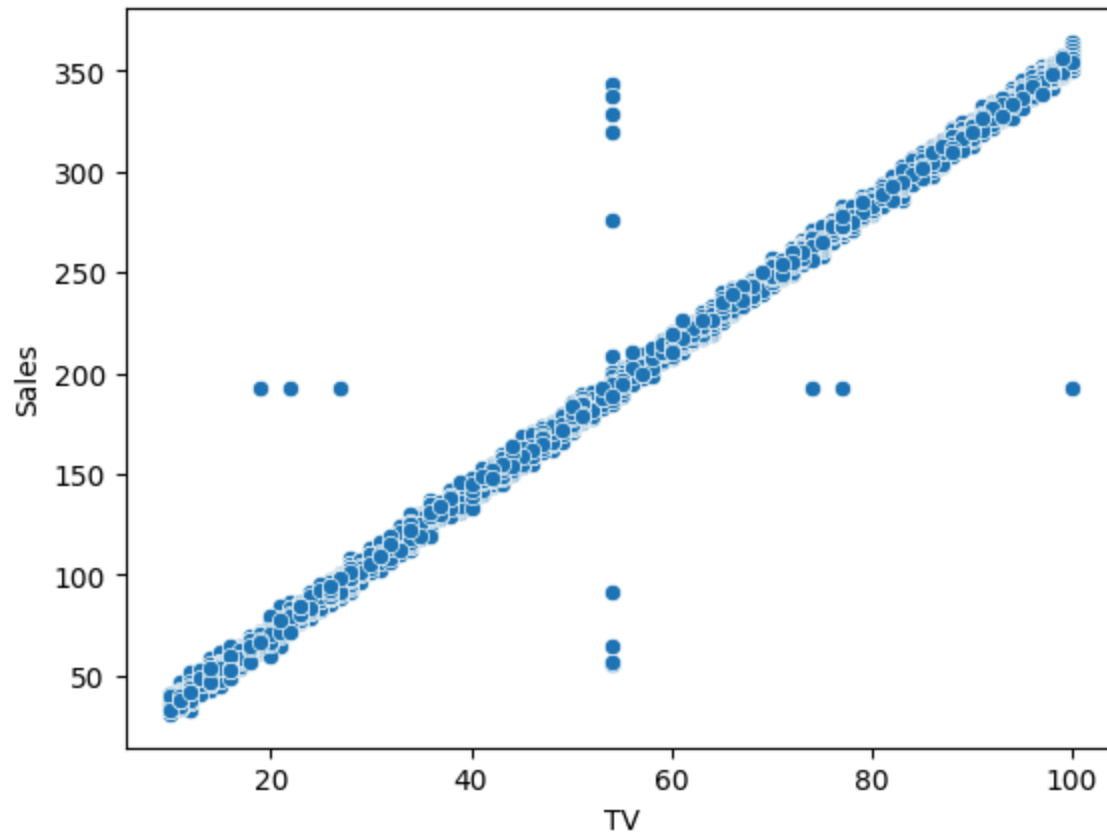
Radio is moderately distributed but contains a few outliers that may influence the model.

Social media has very low variability, suggesting it may have limited impact on sales prediction.

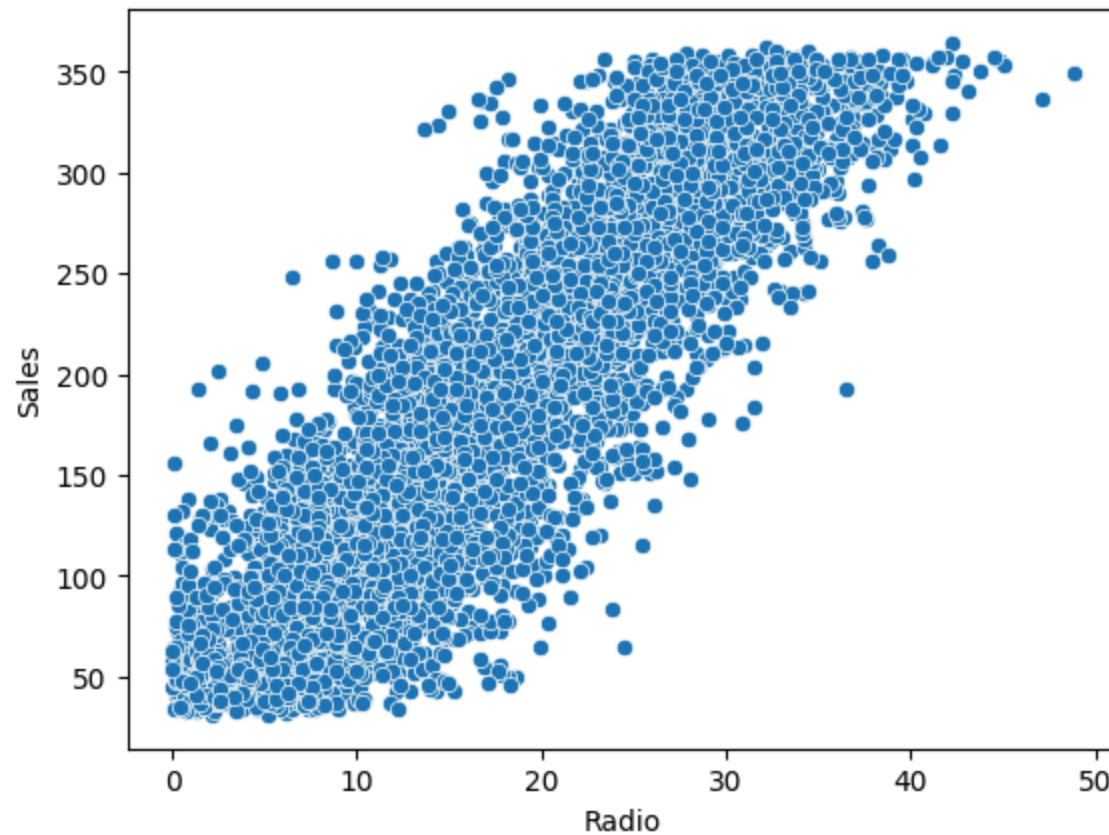
Sales data is widely distributed, indicating sufficient variation for regression analysis.

Bivariate Analysis

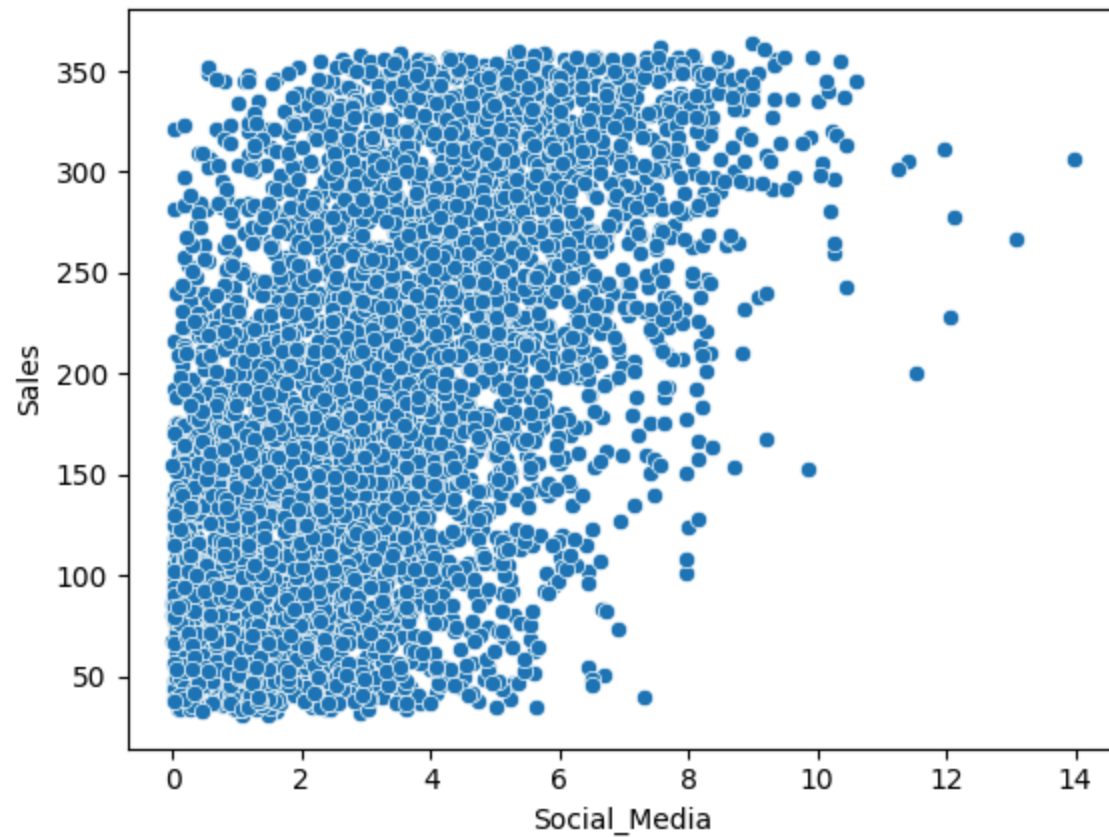
```
sns.scatterplot(x='TV', y='Sales', data=df_clean)  
plt.show()
```




```
sns.scatterplot(x='Radio', y='Sales', data=df_clean)  
plt.show()
```



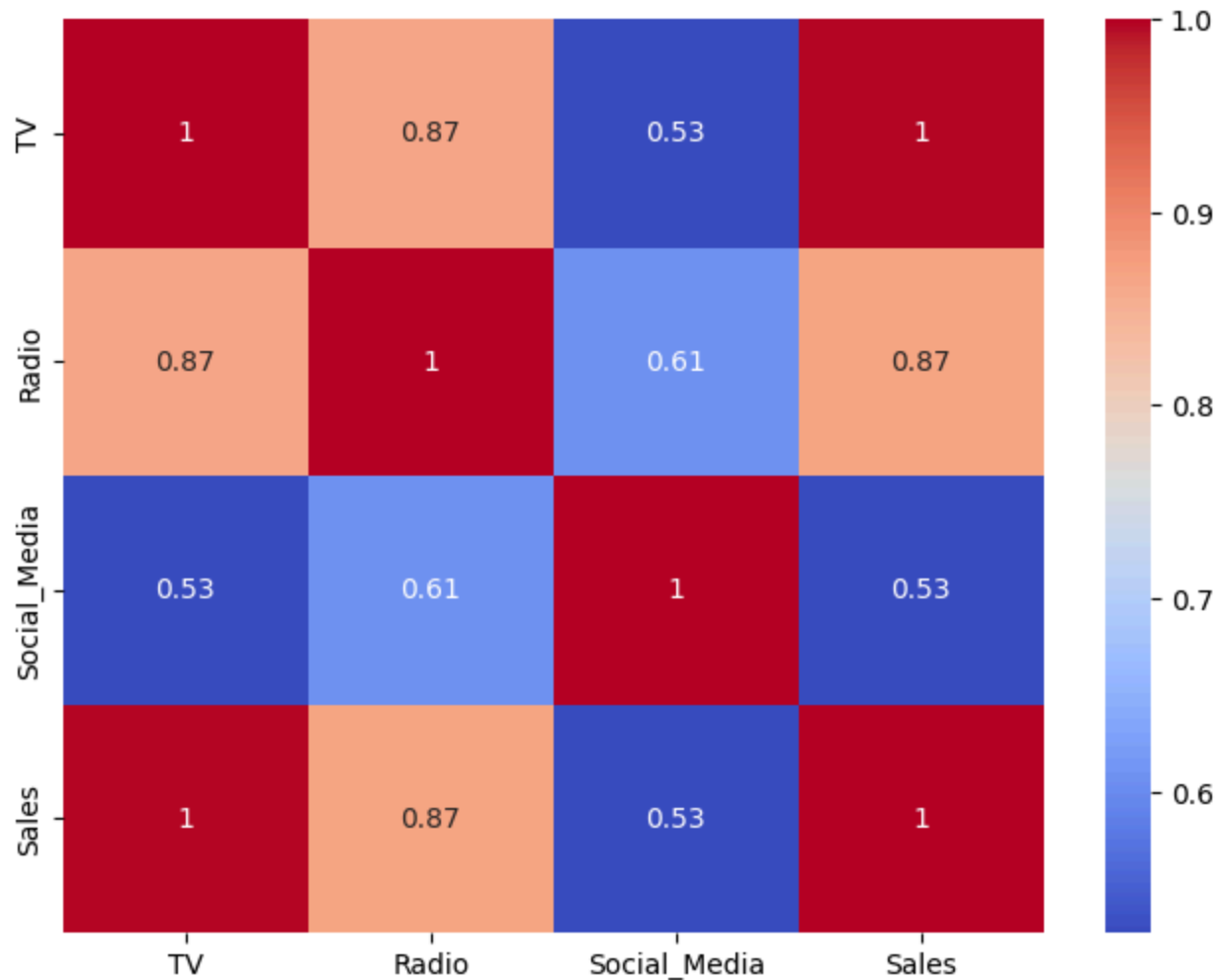
```
sns.scatterplot(x='Social_Media', y='Sales', data=df_clean)  
plt.show()
```



Using the Correlation matrix to identify the Independent variable

```
corr = df_clean.corr()

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.show()
```



Interpretation of the correlation matrix to Identify the Independent variable

1. TV vs Sales $\rightarrow$ 1.00 Perfect relationship TV alone explains 100% of Sales variation

Conclusion:

TV is the BEST possible predictor(Independent variable)

2. Radio vs Sales $\rightarrow$ 0.87 Strong positive correlation

Meaning:

Radio significantly affects Sales But not as perfectly as TV

3. Social Media vs Sales → 0.53 Moderate correlation

Meaning:

Some relationship But much weaker → not ideal for prediction

Final Conclusion The correlation analysis shows that TV advertising as one of the marketing channels has a perfect positive correlation ($r = 1.0$) with sales, indicating a very strong linear relationship. Radio also demonstrates a strong correlation ($r = 0.87$), while social media shows a moderate relationship ($r = 0.53$). Based on these results, TV is selected as the independent variable for the regression model.

Building an OLS Regression model using statmodels

```
#import library
import statsmodels.api as sm
```

```
X = df_clean[['TV']] # independent variable
y = df_clean['Sales'] # dependent variable
#Add constant
X = sm.add_constant(X)
```

```
model = sm.OLS(y, X).fit()
```

View Results

```
print(model.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:              Sales    R-squared:                0.993

```

```

Model:                OLS      Adj. R-squared:      0.993
Method:              Least Squares  F-statistic:      6.792e+05
Date:                Wed, 10 Jun 2026  Prob (F-statistic):      0.00
Time:                11:39:48    Log-Likelihood:      -15765.
No. Observations:    4572      AIC:                3.153e+04
Df Residuals:        4570      BIC:                3.155e+04
Df Model:            1
Covariance Type:      nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const          0.2849      0.259         1.100      0.271      -0.223      0.793
TV             3.5545      0.004       824.111      0.000        3.546      3.563
=====
Omnibus:                2844.142    Durbin-Watson:                2.009
Prob(Omnibus):           0.000    Jarque-Bera (JB):        12214106.398
Skew:                    1.223    Prob(JB):                 0.00
Kurtosis:                256.200    Cond. No.                 138.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression model

Sales = 0.2849 + 3.5545 x TV

1. R-squared = 0.993

This means:

99.3% of the variation in Sales is explained by TV advertising

Interpretation:

The model has an excellent fit, indicating that TV advertising is a very strong predictor of sales.

2. TV Coefficient = 3.5545

Meaning:

For every 1 unit increase in TV advertising, Sales increase by 3.55 units

There is a strong positive relationship between TV advertising and sales, with sales increasing significantly as TV spending rises

3. Intercept (const) = 0.2849 Very small P-value = 0.271 (> 0.05) $\rightarrow$ not significant

Meaning:

When TV = 0, Sales $\approx$ 0.28 (not very meaningful)

The intercept is not statistically significant, suggesting limited practical interpretation.

4. P-value (TV) = 0.000

Meaning:

The relationship between TV and Sales is statistically significant

5. F-statistic = 6.79e+05

Very large $\rightarrow$ model is highly significant overall

Final Conclusion

The OLS regression model shows that TV advertising has a strong and statistically significant effect on sales. The model explains 99.3% of the variation in sales, indicating an excellent fit. The coefficient suggests that for every unit increase in TV advertising, sales increase by approximately 3.55 units. Therefore, TV advertising is the most effective predictor of sales among the variables considered

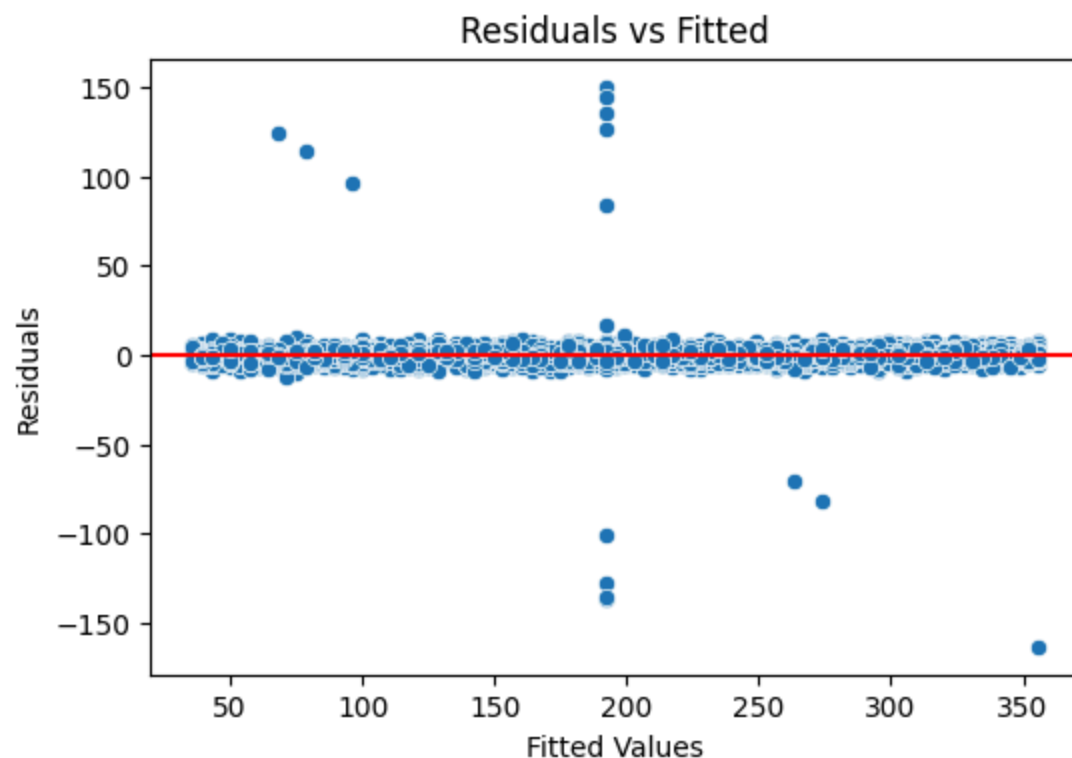
Create diagnostic plots to test Linearity, Normality, and Homoscedasticity

```
#Get residual and fitted value  
fitted_vals = model.fittedvalues  
residuals = model.resid
```

Test Linearity

```
#Use Residuals vs Fitted Plot
```

```
plt.figure(figsize=(6,4))  
sns.scatterplot(x=fitted_vals, y=residuals)  
plt.axhline(0, color='red')  
plt.xlabel("Fitted Values")  
plt.ylabel("Residuals")  
plt.title("Residuals vs Fitted")  
plt.show()
```

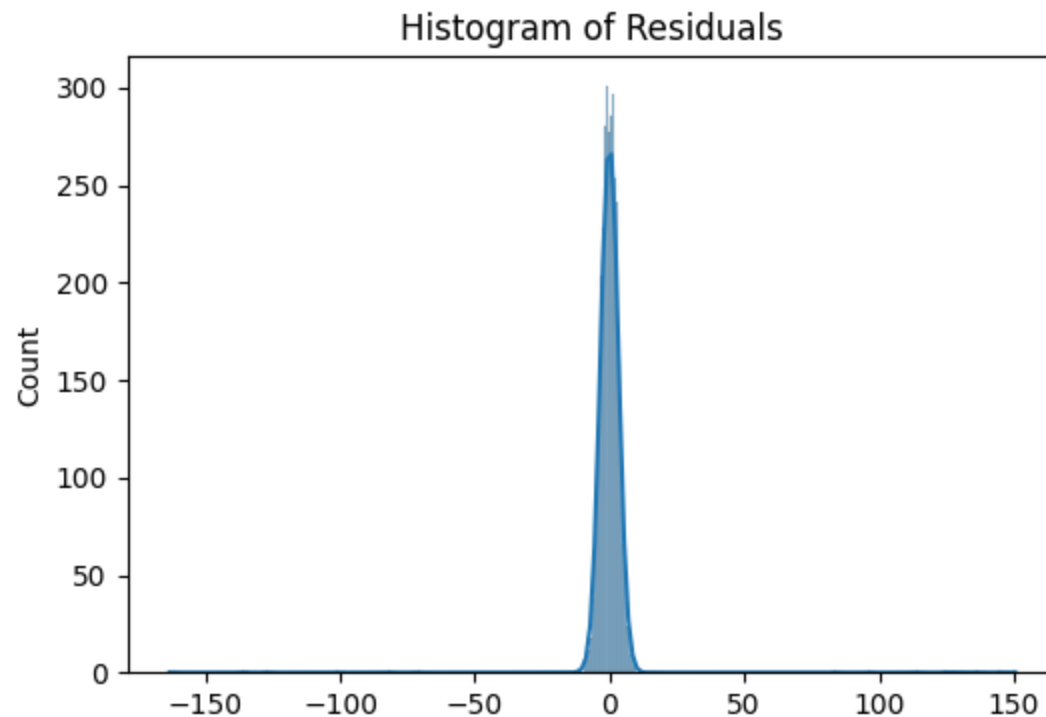


The residuals vs fitted plot shows that the majority of residuals are concentrated along a horizontal line around zero, indicating a strong linear relationship and good model fit. A small number of points display slight vertical and diagonal dispersion, suggesting minor variability and potential localized deviations. However, no clear systematic pattern or curvature is observed, and therefore the linearity assumption is considered to be satisfied.

Test Normality of residuals

(a) Histogram

```
plt.figure(figsize=(6,4))  
sns.histplot(residuals, kde=True)  
plt.title("Histogram of Residuals")  
plt.show()
```

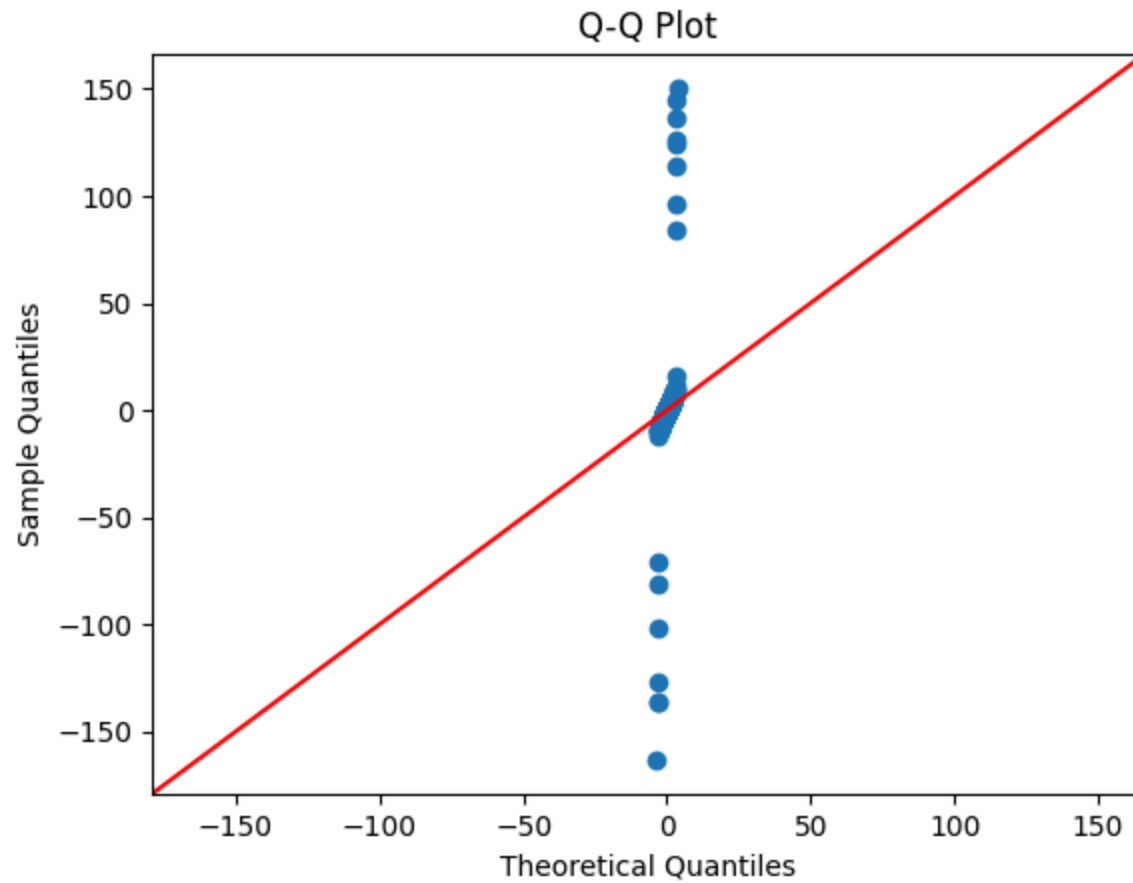


The histogram of residuals shows a sharp peak around zero with the presence of extreme values in the tails. This indicates that the residuals are not normally distributed, exhibiting a leptokurtic (too many values near zero) distribution with heavy tails (extreme outliers).

(b) Q-Q Plot


```
import statsmodels.api as sm

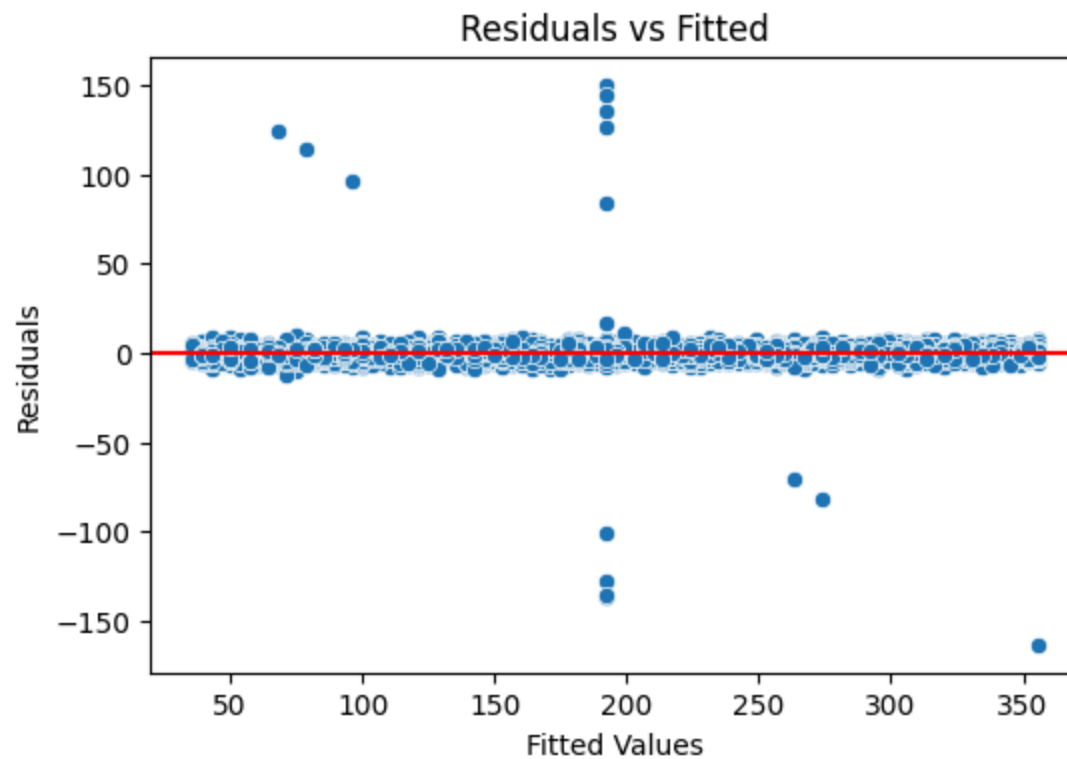
sm.qqplot(residuals, line='45')
plt.title("Q-Q Plot")
plt.show()
```



The Q-Q plot shows that the residuals deviate significantly from the 45-degree reference line, indicating that the assumption of normality is violated. The clustering of points around the center and the presence of extreme deviations at the tails suggest a leptokurtic distribution with heavy tails, likely caused by outliers.

Test Homoscedascity

```
plt.figure(figsize=(6,4))
sns.scatterplot(x=fitted_vals, y=residuals)
plt.axhline(0, color='red')
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.title("Residuals vs Fitted")
plt.show()
```



The residuals vs fitted plot shows that the residuals are randomly scattered around zero with no clear pattern, suggesting that the assumption of homoscedasticity is reasonably satisfied. However, the presence of a few extreme residuals indicates slight deviations from constant variance.

Interpretation of R-squared, coefficients, and p-values in business context

1. R-squared (0.993)

it means 99.3% of the variation in Sales is explained by TV advertising

Business interpretation:

TV advertising is an extremely strong predictor of sales performance. Nearly all changes in sales can be explained by changes in TV advertising spend.

2. Coefficients

From the model:

Intercept (const) = 0.2849

TV coefficient = 3.5545

TV Coefficient (MOST important)

**** What it means:****

- For every 1 unit increase in TV advertising, Sales increase by ~3.55 units

Business interpretation:

Increasing TV advertising expenditure leads to a significant increase in sales. Specifically, each additional unit of TV advertising is associated with an average increase of about 3.55 units in sales

Real meaning: TV advertising is a highly effective marketing channel, delivering strong ROI. Increasing the TV budget is likely to significantly boost revenue

Intercept (0.2849)

What it means

Expected Sales when TV = 0

Business interpretation:

When no money is spent on TV advertising, baseline sales are approximately 0.28 units

p-value = 0.271 → NOT significant So this baseline is not reliable

The intercept is not statistically significant, suggesting that baseline sales without TV advertising may not be meaningful in this model.

3.P-values

From the output:

TV p-value = 0.000 Intercept p-value = 0.271

TV Variable What it means:

p-value: < 0.05 → statistically significant

**** Business interpretation:****

TV advertising has a statistically significant impact on sales, meaning the relationship observed is highly unlikely to be due to chance.

**** Translation**** This is a real effect, not random You can confidently base decisions on it

Intercept

What it means:

p-value=0.271 > 0.05 → NOT significant

Business interpretation:

The intercept is not statistically significant, indicating that the estimated baseline level of sales (when TV advertising is zero) is not reliable.

Final interpretation

The regression model shows a very strong relationship between TV advertising and sales, with an R^2 of 0.993, indicating that 99.3% of the variation in sales is explained by TV advertising. The coefficient for TV advertising (3.5545) suggests that for every unit increase in TV advertising expenditure, sales increase by approximately 3.55 units, highlighting a strong

positive impact. The p-value for TV advertising is less than 0.05, confirming that this relationship is statistically significant and not due to random chance. However, the intercept is not statistically significant, suggesting that the baseline level of sales in the absence of TV advertising is not reliable. Overall, the results indicate that TV advertising is a highly effective driver of sales

ROI Based Recommendation

Core Insight from the Model

- Every 1 unit of TV advertising → ~3.55 units increase in Sales
- Very strong relationship ($R^2 \approx 0.993$)
- Statistically significant ($p < 0.05$)

What This Means for ROI

ROI $\approx$ 3.55 : 1

In plain terms: